STRONG SPLITTING EXISTS AFTER AN ETALE LOCALIZATION

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers within this span; the construction itself lives in the uncarded finite-part lemma and is declared external, never re-proven here.

Allowed standard prerequisites (editorial).

- The universal finite part of a groupoid. lemma-finite-part-groupoid (same chapter, NOT carded, declared external). Supplies the canonical morphism (U', Z_univ, s', t', c') -> (U, R, s, t, c) used as the construction.
- The finite-part section construction. section-finite-part-groupoid of this chapter (same chapter, structural anchor). How (u, Z_u) defines a Spec(kappa(u))-valued point u' of U'.

Source reference. Original lemma 04RU: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2356–2356. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 04RT: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2256–2256. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04RL: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2463–2463. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1 (Existence of strong splitting)**.** *In Situation 04RL there exists an algebraic space U', an étale morphism $U' \to U$, and a point $u' : \mathrm{Spec}(\kappa(u)) \to U'$ lying over $u : \mathrm{Spec}(\kappa(u)) \to U$ such that the restriction $R' = R|_{U'}$ of R to U' is strongly split over u'.*

Proof. Let $f : (U', Z_{univ}, s', t', c') \to (U, R, s, t, c)$ be as constructed in Lemma 04RU. Recall that $R' = R \times_{(U \times_S U)} (U' \times_S U')$. Thus we get a morphism $(f, t', s') : Z_{univ} \to R'$ of groupoids in algebraic spaces

$$(U', Z_{univ}, s', t', c') \to (U', R', s', t', c')$$

Date: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(by abuse of notation we indicate the morphisms in the two groupoids by the same symbols). Now, as $Z_{univ} \subset R \times_{s,U,g} U'$ is open and $R' \to R \times_{s,U,g} U'$ is étale (as a base change of $U' \to U$) we see that $Z_{univ} \to R'$ is an open immersion. By construction the morphisms $s', t' : Z_{univ} \to U'$ are finite. It remains to find the point u' of U'.

We think of u as a morphism $\mathrm{Spec}(\kappa(u)) \to U$ as in the statement of the lemma. Set $F_u = R \times_{s,U} \mathrm{Spec}(\kappa(u))$. The set $\{r \in R : s(r) = u, t(r) = u\}$ is finite by assumption and $F_u \to \mathrm{Spec}(\kappa(u))$ is quasi-finite at each of its elements by assumption. Hence we can find a decomposition into open and closed subschemes

$$F_u = Z_u \amalg Rest$$

for some scheme Z_u finite over $\kappa(u)$ whose support is $\{r \in R : s(r) = u, t(r) = u\}$. Note that $e(u) \in Z_u$. Hence by the construction of U' in Section 04RT (u, Z_u) defines a $\mathrm{Spec}(\kappa(u))$-valued point u' of U'.

We still have to show that the set $\{r' \in |R'| : s'(r') = u', t'(r') = u'\}$ is contained in $|Z_{univ}|$. Pick any point r' in this set and represent it by a morphism $z' : \mathrm{Spec}(k) \to R'$. Denote $z : \mathrm{Spec}(k) \to R$ the composition of z' with the map $R' \to R$. Clearly, z defines an element of the set $\{r \in R : s(r) = u, t(r) = u\}$. Also, the compositions $s \circ z, t \circ z : \mathrm{Spec}(k) \to U$ factor through u, so we may think of $s \circ z, t \circ z$ as a morphism $\mathrm{Spec}(k) \to \mathrm{Spec}(\kappa(u))$. Then $z' = (z, u' \circ t \circ z, u' \circ s \circ u)$ as morphisms into $R' = R \times_{(U \times_S U)} (U' \times_S U')$. Consider the triple

$$(s \circ z, Z_u \times_{\mathrm{Spec}(\kappa(u)), s \circ z} \mathrm{Spec}(k), z)$$

where Z_u is as above. This defines a $\mathrm{Spec}(k)$-valued point of Z_{univ} whose image via s', t' in U' is u' and whose image via $Z_{univ} \to R'$ is the point r' by the relationship between z and z' mentioned above. This finishes the proof. $\square$